General Use SOP

Flammable and Combustible Liquids

Process or Experiment Description

This standard operating procedure (SOP) provides general guidance on working safely with lab-scale quantities of flammable/combustible liquids. In some cases, multiple SOPs may apply to a single chemical (e.g., for benzene, both General Use SOPs for flammable liquids and reproductive toxins would apply). For questions about the applicability of any item in this SOP, contact your Principal Investigator/Supervisor or EH&S at 650-723-0448.

For work with large quantities of flammable materials, complex procedures, or otherwise particularly hazardous procedures, EH&S recommends completing a risk assessment, reviewing other EH&S guidance (such as guidance on scale-up), consulting similar SOPs in the SOP Library, and/or writing a dedicated SOP and submitting it for EH&S review.

Hazards

Flammable and combustible liquids both pose fire hazards. Flammable liquids have a flashpoint below 100 °F (37.8 °C) and combustible liquids have a flash point between 100 °F (37.8 °C) and 200 °F (93.4 °C). The lower a material's flashpoint, the more flammable it is. Other properties, like vapor pressure and explosive limits, also impact the hazard level.

Certain conditions increase the risks associated with these liquids. Flammable and combustible liquids are incompatible with oxidizing materials (e.g., bleach, hydrogen peroxide, nitric acid) and will react, ignite, or, in some cases, even explode. Aerosolizing these liquids increases the risk due to the increased surface area of the droplets, making ignition more likely. Additionally, their high vapor pressures allow flammable vapors to spread and many of these vapors are heavier than air, causing them to sink towards the floor.

Common flammable and combustible liquids include methanol, ethanol, isopropanol, and glacial acetic acid. Many organic solvents are flammable liquids.

Control of Hazards

General

- Minimize the potential for splash, splatter, or other likely scenarios for accidental contact.
    - Use less flammable chemicals, as feasible.
    - Use the smallest quantity feasible.
- Avoid ignition sources.
    - Do not heat flammable or combustible liquids with an open flame and do not use them near an open flame.
    - Do not use electrical devices with cracked or frayed electrical wiring, such as hot plates or heating mantles.
    - Designate work areas for open flames and do not use flammable/combustible liquids in those areas.
- For highly flammable liquids, avoid static electricity or hot surfaces which can be ignition sources.
- Do not use flammable or combustible liquids near strong oxidizers (e.g., 30% hydrogen peroxide). If you are combining these materials as part of an experiment, exercise caution and consult with EH&S before proceeding.
- Remove unnecessary combustible materials (e.g., Kim wipes, plastics, bench pads) from your work area before handling flammable or combustible liquids.
- When transferring flammable liquid from a metal bulk container (greater than five gallons), the container must be electrically bonded and grounded.
- Transfer flammable liquids inside a laboratory hood (or other area with similar ventilation) to prevent vapor accumulation.
- Use a cold trap with rotary evaporators to prevent vapors from reaching the pump.

Engineering/Ventilation Controls

Use a properly functioning chemical fume hood when handling flammable or combustible liquids whenever possible, especially when using >500 mL, or when heating or pressurizing. If the process doesn't allow working in the fume hood, contact Environmental Health and Safety at 650-723-0448.

Personal Protective Equipment

In addition to proper street clothing (long pants or equivalent that covers legs and ankles, and close-toed non-perforated shoes that completely cover the feet), wear the following Personal Protective Equipment (PPE):

- Safety glasses (if splash potential exists, wear goggles and a face shield instead)
- Lab coat (a flame-resistant lab coat for work with >4 liters or based on task, e.g., heating)
- Appropriate chemical-resistant gloves
- Non-synthetic street clothing

Special Handling Procedures and Storage Requirements

When storing 10 gallons or more of flammable/combustible liquids, using a flammable storage cabinet is required. While smaller quantities may not require such storage, it is still recommended. Quantities less than 500 mL (e.g., wash bottles) may be stored in secondary containment on benchtops. Never store these liquids in a refrigerator not rated for flammable material storage.

Appropriate fire extinguishers must be available in all laboratories and storage areas.

Containers should be closed whenever they are not in use. Segregate incompatible chemicals and use secondary containment. Follow any substance-specific storage guidance provided in SDS. Keep flammable and combustible liquids separated from oxidizing materials (e.g., bleach, hydrogen peroxide, nitric acid).

Note that some materials (e.g., glacial acetic acid, peracetic acid) are flammable but are not in Storage Group L. Such materials should be stored in separate secondary containment within a flammables cabinet.

Emergency Procedures

All incidents should be reported to EH&S with an incident report form.

Spill

Refer to EH&S' guidance on hazardous materials incidents for general procedures for incident response, and call 650-725-9999 for assistance with spill cleanup.

If you are trained in spill cleanup and the spill is not health-threatening, proceed with spill cleanup. There may be variations in procedure based on the specific spill, but the general steps are:

1. Don the appropriate PPE (See Personal Protective Equipment section, above).
2. Locate the nearest spill kit and bring it to the spill.
3. Use the absorbent pads in the spill kit to wipe up the spill.
4. Place the used absorbent pad and gloves inside the spill kit jar.
5. Place a waste tag on the spill kit and request waste pickup.

Be aware of possible ignition sources. Spills increase the surface area of the liquid, causing it to vaporize more quickly, which may lead to respiratory irritation, inhalation of toxic vapors, and/or a flammable atmosphere.

Exposure

If immediate medical attention is required, call 911. Remove any contaminated clothing, and IMMEDIATELY flush contaminated skin with water for at least 15 minutes. Use a safety shower for any skin exposures to the trunk, legs, or feet.

For eye exposures, IMMEDIATELY flush eyes with water for at least 15 minutes using an eyewash station.

If medical attention is needed, provide the SDS(s) of the chemical(s) to aid medical staff in proper diagnosis and treatment.

For minor exposures, contact the Stanford University Occupational Health Center at 650-725-5308. For serious exposures, go to the Stanford Hospital Emergency Department.

Those who work with hazardous chemicals are entitled to receive medical attention/consultation when:

- A spill, leak, explosion or other occurrence results in a hazardous exposure (potential overexposure)
- Symptoms or signs of exposure to a hazardous chemical develop

Fire

If flammable/combustible liquids catch on fire, it may be extinguished with a CO_2 extinguisher if the fire is small (smaller than a trash can) and you've been trained. Remember PASS:

Pull the pin

Aim at the base of the fire

Squeeze the nozzle

Sweep back and forth

Use the extinguisher from 6 or more feet away. Using the extinguisher too close to the fire risks spraying the ignited liquid.

Waste Disposal

Dispose of flammable and combustible liquids as hazardous waste. All hazardous waste containers must have a waste tag attached before any waste is added. Before combining materials in waste containers, review the SDSs and Stanford's Incompatibility Guide. For general guidance on waste disposal, refer to the guide for handling and storing waste and the chemical waste poster.

Minimum Training Requirements

- General Safety & Emergency Preparedness (EHS-4200)
- Chemical Safety for Laboratories (EHS-1900)
- Laboratory-specific training
- Fire Extinguisher Use (EHS 3700) – recommended

Approval Required

Consult with your PI regarding the need for prior approval. The PI/Lab Supervisor shall provide prior approval of any chemical usage involving Restricted Chemicals or High Risk Procedures.

Designated Area

For flammable or combustible liquids that are also considered particularly hazardous substances (i.e., select carcinogens, reproductive toxins, and highly acutely toxic materials), a designated area shall be established per the other applicable SOP(s).